A-level

ENVIRONMENTAL SCIENCE

Questions by topic: The circular economy, biocapacity & ecological footprints - Set 1

Mark scheme

AQA spec 3.6.4; Sustainability; textbook pp. 374-389

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 6 points below. **1** Cycling of materials (like biogeochemical cycles) - recycling, re-use **1** Energy from renewable sources **1** Human activities should support ecosystems / conserve biodiversity **1** Separation of technical and biological materials – connected systems where the waste of one process feeds another (industrial symbiosis) **1** Diverse systems are more resistant to change **1** Design of products for extended use and end-of-life re-use – optimum rather than maximum production – technologies that improve efficiency and enable dismantling **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new • **biodiversity** → the variety of life: habitat, species and genetic diversity • **symbiosis** → a close relationship in which both species benefit **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	6

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae **1** Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them **1** A common approach gives a level playing field, coordinated monitoring and support for less successful countries **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	

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Question	Marking guidance	Marks
02.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

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Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
02.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1 Ecological footprint:** the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced **1 Biocapacity:** the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** A hectare of land / sea with the world-average biological productivity **1** Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 8.1 / 1.7 = 4.8 (accept 4.7-4.8) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (Earths) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1

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Question	Marking guidance	Marks
03.4	**1 mark each** 1 mark for each of the 3 points below. **1** Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint **1** Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO2 accumulates (overshoot) **1** Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up – it falls earlier each year (8 August in 2016) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Renewable energy and energy conservation (carbon footprint is the largest component) **1** Reduced meat consumption / food waste – local seasonal food **1 Circular economy:** repair, re-use, recycling – longer product lives **1** Public transport, efficient buildings **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **energy conservation** → using less energy, eg by insulation or behaviour change • **recycling** → reprocessing a material to make something new **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** All greenhouse gases expressed as CO2 equivalent (allowing for different warming strengths) **1** Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) **1** A fair share of the infrastructure: roads, car parks, signals, street lighting **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

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Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels **1** Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
04.3	**1 mark each** 1 mark for each of the 5 points below. **1** Hardy, disease-resistant Boran cattle – regular tick dipping reduces ticks for wild herbivores too **1** Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth **1** Electric fences keep game in and poachers out – biological corridors let animals (except rhinos) move to other reserves – problem elephants relocated **1 Monitoring:** radio-collared lions, drones, camera traps, track surveys – endangered species (Grevy's zebra, northern white rhino) protected **1 Community links:** ecotourism income funds health, education and water projects – local employment – cattle breeding help for local farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
05.1	**1 mark each** 1 mark for each of the 3 points below. **1 Diversity:** HEP (wet winters), solar (summer) and wind peak at different times so supply is more reliable - diverse systems resist change **1** No single source could meet demand all year – battery storage on the island grid smooths fluctuations **1** Calm, dry, dull periods still leave gaps so diesel generators provide security (fossil fuel use minimised, not yet eliminated) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	3

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Question	Marking guidance	Marks
05.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Community ownership (Isle of Eigg Trust with Scottish Wildlife Trust) managing the island for people and wildlife **1** Wood biomass heating – insulation such as straw-bale housing – some electric vehicles **1 Habitat protection and creation:** fencing rare plants from grazing, planting indigenous woodland (rowan, birch, willow) – eagles and hen harriers nesting **1** Recyclables shipped off – garden / food waste composted – low-impact food production – beach litter collection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainability** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Small groups can organise, agree and act quickly – decisions benefit people who see the results directly **1** Resources and wastes are local and visible so feedback is immediate, unlike cities supplied from far away **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Renewable energy is abundant but geographically and temporally variable – a continental grid moves electricity from areas of surplus (desert solar, offshore wind) to areas of demand **1** DC grids do not need every generator synchronised in phase as AC grids do, so coordination across many countries is easier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 384-389; AQA spec 3.6.4.	

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